

COVID19 DATA

FRANK CARSONIE

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PREFACE

I reused much of the code that Dr. Jane Wall presented in her lectures with the COVID 19 data for this final project, and have added comments, additional analysis and visualizations, and an additional model to the data for review.

The data presented is a COVID19 dataset from the Johns Hopkins Github site presented in the URL below.

INTRODUCTION

Began by loading all the library packages needed for the data manipulation.

```
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
## v dplyr      1.1.4      v readr      2.1.5
## v forcats    1.0.0      v stringr   1.5.2
## v ggplot2    4.0.0      v tibble    3.3.0
## v lubridate  1.9.4      v tidyr     1.3.1
## v purrr      1.1.0
## -- Conflicts ----- tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(lubridate)
library(dplyr)
library(modelr)
library(knitr)
```

URL LOADING

Gathered the URL data from the COVID19 master data, with additional confirmed global cases, confirmed global deaths, confirmed US cases, and confirmed US deaths as additional csv's, loaded into a table of url strings.

```
url_in <- "https://raw.githubusercontent.com/CSSEGISandData/COVID-19/master/csse_covid_19_data/csse_cov
file_names <- c("time_series_covid19_confirmed_global.csv", "time_series_covid19_deaths_global.csv", "t
urls<- str_c(url_in,file_names)
```

DATA VARIABLE CREATION

Read the csv data from each of the url strings into specific variables, titles global_cases, global_deaths, US_cases, and US_deaths

```
global_cases <- read_csv(urls[1])
global_deaths <- read_csv(urls[2])
US_cases <- read_csv(urls[3])
US_deaths <- read_csv(urls[4])
```

GLOBAL DATA CLEAN UP

Pivoted the table information for global cases and global deaths via Province/State, Country/Region, and Lat, Long information so they would be on a single row, and cleaned up variable types and removed unnecessary columns.

```
global_cases <- global_cases %>%
  pivot_longer(cols = -c('Province/State', 'Country/Region', Lat, Long),
    names_to = "date",
    values_to = "cases") %>%
  select(-c(Lat,Long))

global_deaths <- global_deaths %>%
  pivot_longer(cols = -c('Province/State', 'Country/Region', Lat, Long),
    names_to = "date",
    values_to = "deaths") %>%
  select(-c(Lat,Long))
```

COMBINE GLOBAL DATA

Joined the global cases and global deaths tables into one table for ease of manipulation, and changed the date data type to date

```
global <- global_cases %>%
  full_join(global_deaths) %>%
  rename(Country_Region = 'Country/Region', Province_State = 'Province/State') %>%
  mutate(date = mdy(date))
```

```
## Joining with 'by = join_by('Province/State', 'Country/Region', date)'
```

VIEW SUMMARY GLOBAL

Viewed a summary of the global data information and made sure to filter out any days where the cases were less than 1 for brevity

```
summary(global)
```

```
## Province_State      Country_Region      date      cases
## Length:330327      Length:330327      Min.   :2020-01-22      Min.   :      0
## Class :character    Class :character    1st Qu.:2020-11-02      1st Qu.:      680
## Mode  :character    Mode  :character    Median :2021-08-15      Median :     14429
##                                     Mean  :2021-08-15      Mean  :    959384
##                                     3rd Qu.:2022-05-28      3rd Qu.:   228517
##                                     Max.   :2023-03-09      Max.   :103802702
##
##      deaths
## Min.   :      0
## 1st Qu.:      3
## Median :     150
## Mean   :    13380
## 3rd Qu.:    3032
## Max.   :1123836
```

```
global <- global %>% filter(cases>0) # filter out days with less than 1 case
```

US DATA CLEAN UP

Pivoted table information for US cases and US deaths so it would be placed on a single row, and deleted unnecessary lat, long information, while also joining US cases and US deaths into a single variable table titled US

```
US_cases %>%
  pivot_longer(cols = -(UID:Combined_Key), names_to = "date", values_to = "cases")
```

```
## # A tibble: 3,819,906 x 13
##       UID iso2 iso3 code3 FIPS Admin2 Province_State Country_Region Lat
##       <dbl> <chr> <chr> <dbl> <dbl> <chr>   <chr>           <chr>      <dbl>
## 1 84001001 US    USA    840  1001 Autauga Alabama      US        32.5
## 2 84001001 US    USA    840  1001 Autauga Alabama      US        32.5
## 3 84001001 US    USA    840  1001 Autauga Alabama      US        32.5
## 4 84001001 US    USA    840  1001 Autauga Alabama      US        32.5
## 5 84001001 US    USA    840  1001 Autauga Alabama      US        32.5
## 6 84001001 US    USA    840  1001 Autauga Alabama      US        32.5
## 7 84001001 US    USA    840  1001 Autauga Alabama      US        32.5
## 8 84001001 US    USA    840  1001 Autauga Alabama      US        32.5
## 9 84001001 US    USA    840  1001 Autauga Alabama      US        32.5
## 10 84001001 US    USA    840  1001 Autauga Alabama      US        32.5
## # i 3,819,896 more rows
## # i 4 more variables: Long_ <dbl>, Combined_Key <chr>, date <chr>, cases <dbl>
```

```

US_cases <- US_cases %>%
  pivot_longer(cols = -(UID:Combined_Key), names_to = "date", values_to = "cases") %>%
  select(Admin2:cases) %>%
  mutate(date = mdy(date)) %>%
  select(-c(Lat, Long_))

US_deaths <- US_deaths %>%
  pivot_longer(cols = -(UID:Population), names_to = "date", values_to = "deaths") %>%
  select(Admin2:deaths) %>%
  mutate(date = mdy(date)) %>%
  select(-c(Lat, Long_))

US <- US_cases %>% full_join(US_deaths)

```

```

## Joining with 'by = join_by(Admin2, Province_State, Country_Region,
## Combined_Key, date)'

```

COMBINE POPULATION DATA

Combined global population data to create clear population information for each province/state and country/region

```

global <- global %>%
  unite("Combined_Key",
        c(Province_State, Country_Region),
        sep = ", ",
        na.rm = TRUE,
        remove = FALSE)

```

US BY STATE

Grouped US information by state to have a clearer understanding of how COVID19 effected each individual state, gathering the cases, deaths, and population for each state. Additionally, created the US totals to understand how thte US was effected as a whole in terms of deaths per million.

```

US_by_state <- US %>%
  group_by(Province_State, Country_Region, date) %>%
  summarize(cases = sum(cases), deaths = sum(deaths), Population = sum(Population)) %>%
  mutate(deaths_per_mill = deaths *1000000 / Population) %>%
  select(Province_State, Country_Region, date, cases, deaths, deaths_per_mill, Population) %>%
  ungroup()

```

```

## 'summarise()' has grouped output by 'Province_State', 'Country_Region'. You can
## override using the '.groups' argument.

```

```

US_totals <- US %>%
  group_by(Country_Region, date) %>%
  summarize (cases = sum(cases), deaths = sum(deaths), Population = sum(Population)) %>%
  mutate(deaths_per_mill = deaths *1000000 / Population) %>%

```

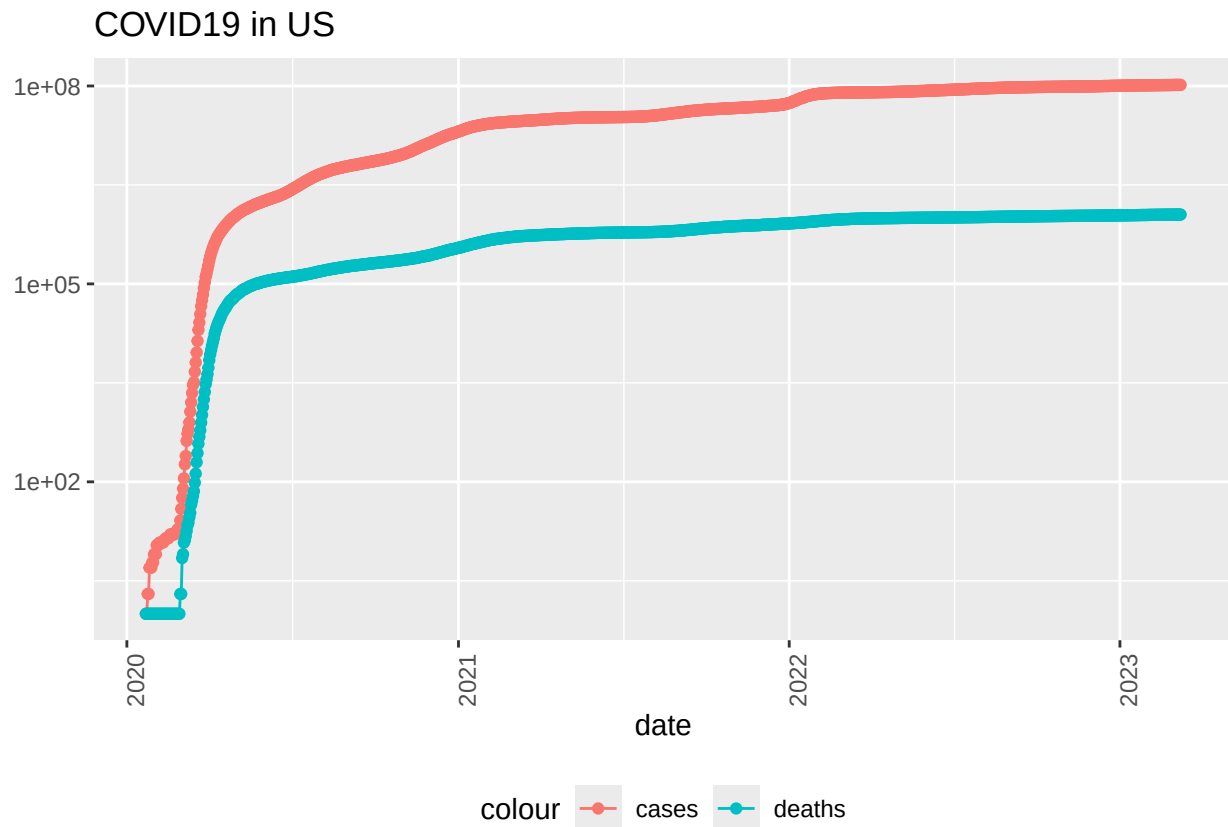
```
select(Country_Region, date, cases, deaths, deaths_per_mill, Population) %>%
ungroup()
```

'summarise()' has grouped output by 'Country_Region'. You can override using
the '.groups' argument.

PLOTTING US COVID CASES AND DEATHS

Used ggplot to plot COVID19 cases and deaths in the US as a whole and in a specific state, in this case New York, to see how they may have differed in the spread of the disease.

```
US_totals %>%
  filter (cases>0) %>%
  ggplot(aes(x = date, y = cases)) +
  geom_line(aes(color = "cases")) +
  geom_point(aes(color = "cases")) +
  geom_line(aes(y = deaths, color = "deaths")) +
  geom_point(aes(y=deaths, color = "deaths")) +
  scale_y_log10() +
  theme(legend.position = "bottom", axis.text.x = element_text(angle = 90)) +
  labs(title = "COVID19 in US", y = NULL)
```



```

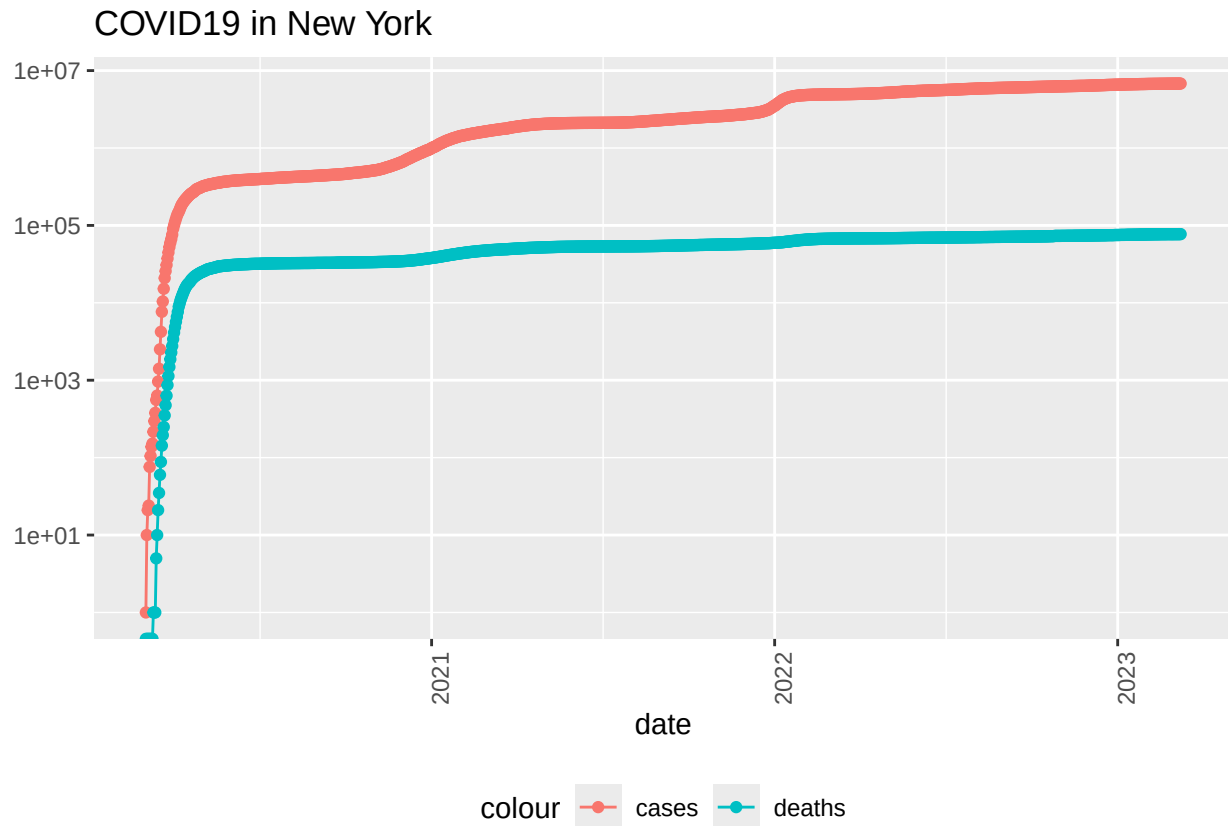
state <- "New York"
US_by_state %>%
  filter (Province_State == state) %>%
  filter (cases > 0) %>%
  ggplot(aes(x = date, y = cases)) +
  geom_line(aes(color = "cases")) +
  geom_point(aes(color = "cases")) +
  geom_line(aes(y = deaths, color = "deaths")) +
  geom_point(aes(y=deaths, color = "deaths")) +
  scale_y_log10() +
  theme(legend.position = "bottom", axis.text.x = element_text(angle = 90)) +
  labs(title = str_c("COVID19 in ", state), y = NULL)

```

```

## Warning in scale_y_log10(): log-10 transformation introduced infinite values.
## log-10 transformation introduced infinite values.

```



US TOTALS ANALYSIS

Found the date that was the last date where data was captured in the US

```
max(US_totals$date)
```

```
## [1] "2023-03-09"
```

ANALYZING NEW US CASES

Defined US State information to analyze the new COVID19 cases and new COVID 19 deaths to see how they were trending new the end of the data collection period

```
US_by_state <- US_by_state %>%
  mutate(new_cases = cases - lag(cases), new_deaths = deaths - lag(deaths))
US_totals <- US_totals %>%
  mutate(new_cases = cases - lag(cases), new_deaths = deaths - lag(deaths))

tail(US_totals %>% select(new_cases, new_deaths, everything()))
```

```
## # A tibble: 6 x 8
##   new_cases new_deaths Country_Region date       cases deaths deaths_per_mill
##   <dbl>      <dbl> <chr>          <date>    <dbl> <dbl>          <dbl>
## 1      2147         7 US            2023-03-04 1.04e8 1.12e6          3371.
## 2     -3862        -38 US            2023-03-05 1.04e8 1.12e6          3371.
## 3      8564         47 US            2023-03-06 1.04e8 1.12e6          3371.
## 4     35371        335 US            2023-03-07 1.04e8 1.12e6          3372.
## 5     64861        730 US            2023-03-08 1.04e8 1.12e6          3374.
## 6     46931        590 US            2023-03-09 1.04e8 1.12e6          3376.
## # i 1 more variable: Population <dbl>
```

PLOTTING NEW US CASES

Plotted the new US cases and deaths for the US as a whole and New York as a state specifically, visualizing how they compared to the total cases previously explored.

```
US_totals %>%
  ggplot(aes(x = date, y = new_cases)) +
  geom_line(aes(color = "new_cases")) +
  geom_point(aes(color = "new_cases")) +
  geom_line(aes(y = new_deaths, color = "new_deaths")) +
  geom_point(aes(y = new_deaths, color = "new_deaths")) +
  scale_y_log10() +
  theme(legend.position = "bottom", axis.text.x = element_text(angle = 90)) +
  labs(title = "COVID19 in US", y = NULL)
```

```
## Warning in transformation$transform(x): NaNs produced

## Warning in scale_y_log10(): log-10 transformation introduced infinite values.

## Warning in transformation$transform(x): NaNs produced

## Warning in scale_y_log10(): log-10 transformation introduced infinite values.

## Warning in transformation$transform(x): NaNs produced

## Warning in scale_y_log10(): log-10 transformation introduced infinite values.
```

```
## Warning in transformation$transform(x): NaNs produced

## Warning in scale_y_log10(): log-10 transformation introduced infinite values.

## Warning: Removed 1 row containing missing values or values outside the scale range
## ('geom_line()').

## Warning: Removed 2 rows containing missing values or values outside the scale range
## ('geom_point()').

## Warning: Removed 1 row containing missing values or values outside the scale range
## ('geom_line()').

## Warning: Removed 4 rows containing missing values or values outside the scale range
## ('geom_point()').
```

COVID19 in US

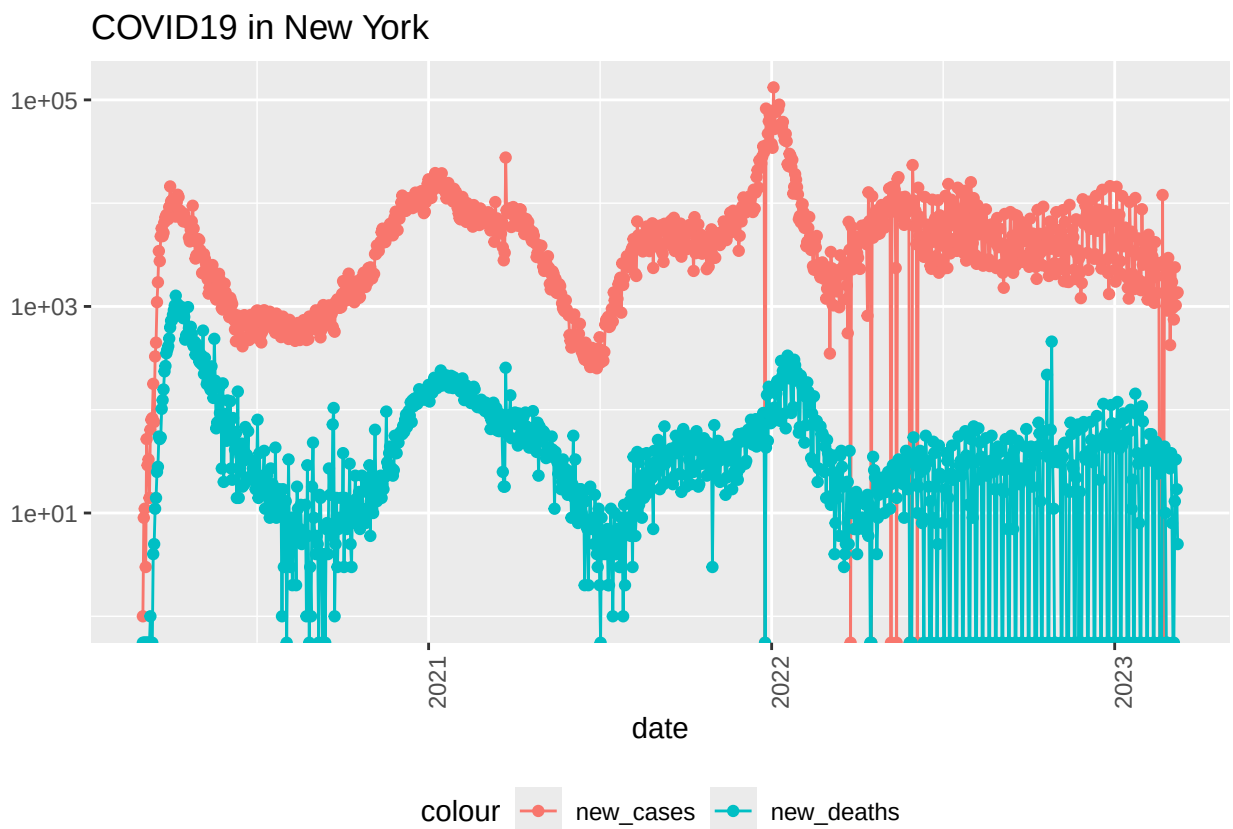


```
state <- "New York"
US_by_state %>%
  filter (Province_State == state) %>%
  filter (cases > 0) %>%
  ggplot(aes(x = date, y = new_cases)) +
  geom_line(aes(color = "new_cases")) +
  geom_point(aes(color = "new_cases")) +
  geom_line(aes(y = new_deaths, color = "new_deaths")) +
```



```
geom_point(aes(y= new_deaths, color = "new_deaths")) +
scale_y_log10() +
theme(legend.position = "bottom", axis.text.x = element_text(angle = 90)) +
labs(title = str_c("COVID19 in ", state), y = NULL)
```

```
## Warning in scale_y_log10(): log-10 transformation introduced infinite values.
## Warning in scale_y_log10(): log-10 transformation introduced infinite values.
## Warning in transformation$transform(x): NaNs produced
## Warning in scale_y_log10(): log-10 transformation introduced infinite values.
## Warning in transformation$transform(x): NaNs produced
## Warning in scale_y_log10(): log-10 transformation introduced infinite values.
## Warning: Removed 8 rows containing missing values or values outside the scale range
## ('geom_point()').
```



ANALYZED DEATHS PER THOUSAND PEOPLE IN EACH STATE

Used the previous information to create a table showing the deaths per thousand for each state in the US. Proceeded to analyze the states with the min and max deaths per thousand to better understand which states were hit the hardest or the least hard in relation to their population density.

```
US_state_totals <- US_by_state %>%
  group_by(Province_State) %>%
  summarize(deaths = max(deaths), cases= max(cases),
            population = max(Population),
            cases_per_thou = 1000 * cases / population,
            deaths_per_thou = 1000 * deaths / population) %>%
  filter (cases > 0, population > 0)

US_state_totals %>%
  slice_min(deaths_per_thou, n=10) %>%
  select(deaths_per_thou, cases_per_thou, everything())
```

```
## # A tibble: 10 x 6
##   deaths_per_thou cases_per_thou Province_State deaths cases population
##   <dbl>          <dbl> <chr>          <dbl> <dbl>    <dbl>
## 1         0.611         150. American Samoa         34 8.32e3    55641
## 2         0.744         248. Northern Mariana Isl~         41 1.37e4    55144
## 3         1.21         231. Virgin Islands         130 2.48e4   107268
## 4         1.30         269. Hawaii         1841 3.81e5   1415872
## 5         1.49         245. Vermont          929 1.53e5    623989
## 6         1.55         293. Puerto Rico        5823 1.10e6   3754939
## 7         1.65         340. Utah          5298 1.09e6   3205958
## 8         2.01         415. Alaska          1486 3.08e5    740995
## 9         2.03         252. District of Columbia        1432 1.78e5    705749
## 10        2.06         253. Washington       15683 1.93e6   7614893
```

```
US_state_totals %>%
  slice_max(deaths_per_thou, n=10) %>%
  select(deaths_per_thou, cases_per_thou, everything())
```

```
## # A tibble: 10 x 6
##   deaths_per_thou cases_per_thou Province_State deaths cases population
##   <dbl>          <dbl> <chr>          <dbl> <dbl>    <dbl>
## 1         4.55         336. Arizona        33102 2443514   7278717
## 2         4.54         326. Oklahoma        17972 1290929   3956971
## 3         4.49         333. Mississippi     13370 990756    2976149
## 4         4.44         359. West Virginia    7960 642760    1792147
## 5         4.32         320. New Mexico       9061 670929    2096829
## 6         4.31         334. Arkansas        13020 1006883   3017804
## 7         4.29         335. Alabama         21032 1644533   4903185
## 8         4.28         368. Tennessee       29263 2515130   6829174
## 9         4.23         307. Michigan        42205 3064125   9986857
## 10        4.06         385. Kentucky        18130 1718471   4467673
```

MODELED

Ran a basic model comparing deaths per thou to cases per thou for US totals, which related the number of deaths to the number of cases. Used this information to create a prediction model relating these two factors, to understand how the number of cases affected the number of deaths. It was rather clear that there tends to be a linear correlation between the two factors.

```
mod <- lm(deaths_per_thou ~ cases_per_thou, data = US_state_totals)
summary(mod)
```

```
##
## Call:
## lm(formula = deaths_per_thou ~ cases_per_thou, data = US_state_totals)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -2.3352 -0.5978  0.1491  0.6535  1.2086
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  -0.36167    0.72480  -0.499    0.62
## cases_per_thou  0.01133    0.00232   4.881 9.76e-06 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.8615 on 54 degrees of freedom
## Multiple R-squared:  0.3061, Adjusted R-squared:  0.2933
## F-statistic: 23.82 on 1 and 54 DF,  p-value: 9.763e-06
```

```
US_state_totals %>% slice_min(cases_per_thou)
```

```
## # A tibble: 1 x 6
##   Province_State deaths cases population cases_per_thou deaths_per_thou
##   <chr>          <dbl> <dbl>      <dbl>          <dbl>          <dbl>
## 1 American Samoa      34  8320      55641          150.           0.611
```

```
US_state_totals %>% slice_max(cases_per_thou)
```

```
## # A tibble: 1 x 6
##   Province_State deaths cases population cases_per_thou deaths_per_thou
##   <chr>          <dbl> <dbl>      <dbl>          <dbl>          <dbl>
## 1 Rhode Island    3870 460697  1059361          435.           3.65
```

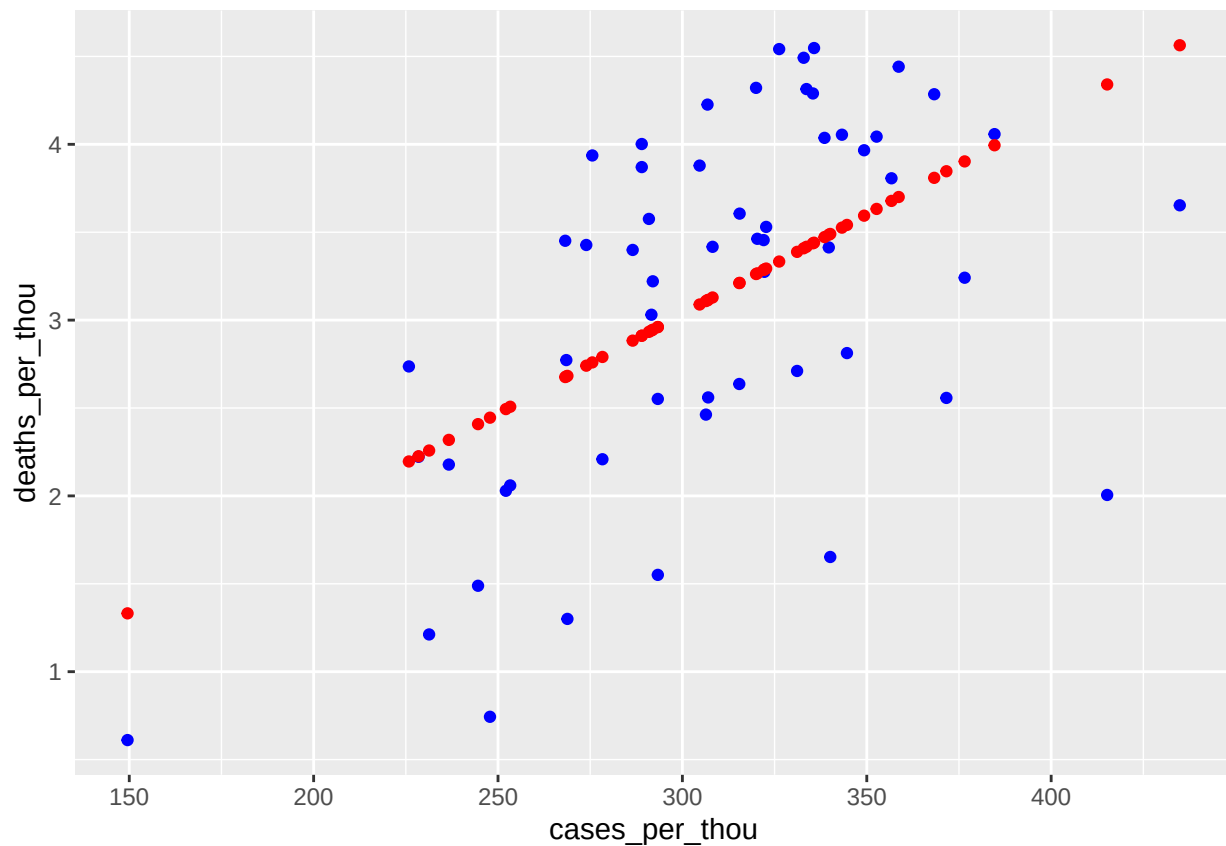
```
US_state_totals %>% mutate(pred = predict(mod))
```

```
## # A tibble: 56 x 7
##   Province_State deaths cases population cases_per_thou deaths_per_thou pred
##   <chr>          <dbl> <dbl>      <dbl>          <dbl>          <dbl> <dbl>
## 1 Alabama      21032 1.64e6  4903185          335.           4.29  3.44
## 2 Alaska        1486 3.08e5   740995          415.           2.01  4.34
## 3 American Samoa      34 8.32e3   55641          150.           0.611 1.33
```

```
## 4 Arizona      33102 2.44e6   7278717      336.      4.55  3.44
## 5 Arkansas     13020 1.01e6   3017804      334.      4.31  3.42
## 6 California  101159 1.21e7  39512223      307.      2.56  3.12
## 7 Colorado     14181 1.76e6   5758736      306.      2.46  3.11
## 8 Connecticut  12220 9.77e5   3565287      274.      3.43  2.74
## 9 Delaware      3324 3.31e5    973764      340.      3.41  3.49
## 10 District of Co~ 1432 1.78e5    705749      252.      2.03  2.49
## # i 46 more rows
```

```
US_tot_w_pred <- US_state_totals %>% mutate(pred = predict(mod))
```

```
US_tot_w_pred %>% ggplot() +
  geom_point(aes(x = cases_per_thou, y = deaths_per_thou), color = "blue") +
  geom_point(aes(x = cases_per_thou, y = pred), color = "red")
```



THE END OF DR. JANE WALL CLASS WORK

This marks the end of the material discussed and worked through with Dr. Jane Walls lectures. From here on, I will include additional visualizations, analysis, and models that interested me regarding the COVID19 data set, and any bias that may have been present in investigations.

ADDITIONAL PERSONAL ANALYSIS

Began by counting the number of cases and number of deaths per year during the COVID19 Pandemic, and joining that into its own data set

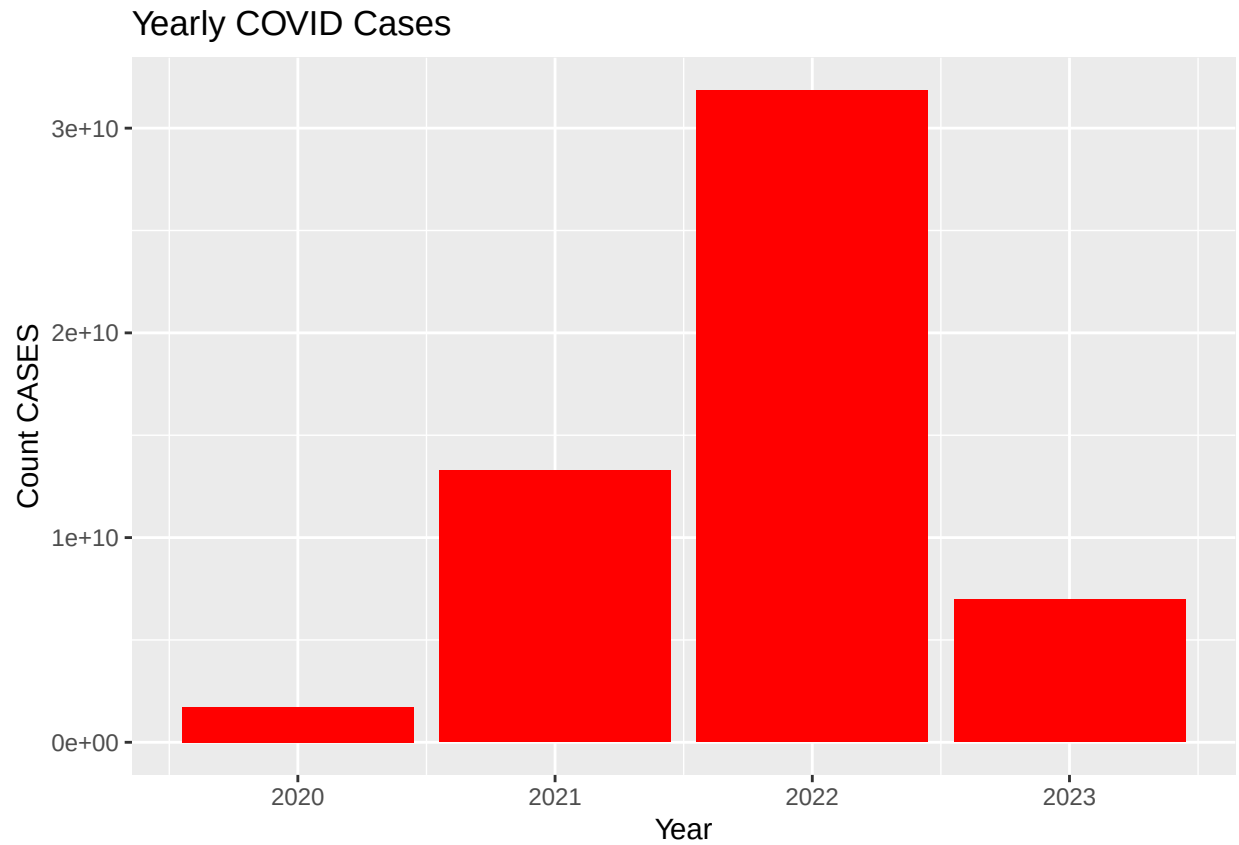
```
yearly_cases <- US_totals %>%  
  group_by(yearly = year(date)) %>%  
  summarize(count_cases = sum(cases)) %>%  
  arrange(yearly)  
  
yearly_deaths <- US_totals %>%  
  group_by(yearly = year(date)) %>%  
  summarize(count_deaths = sum(deaths)) %>%  
  arrange(yearly)  
  
yearly_counts <- yearly_cases %>% full_join(yearly_deaths)
```

```
## Joining with 'by = join_by(yearly)'
```

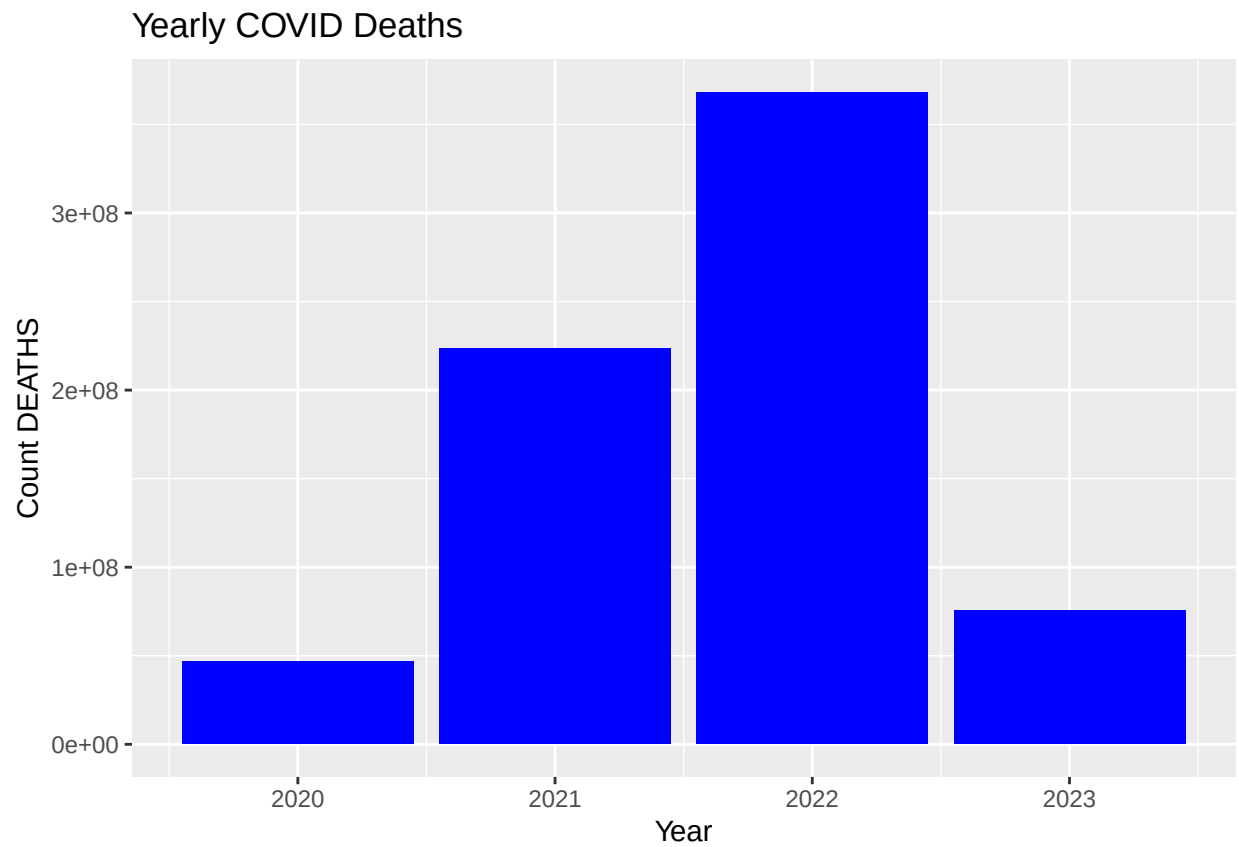
PLOTTING YEARLY DATA

I then plotted the yearly cases and the yearly deaths on different graphs and the same graph to try and get a visualization of the difference. It became clear that the trend of deaths and cases is almost the same in terms of visualization, but they are about 100x more cases than deaths.

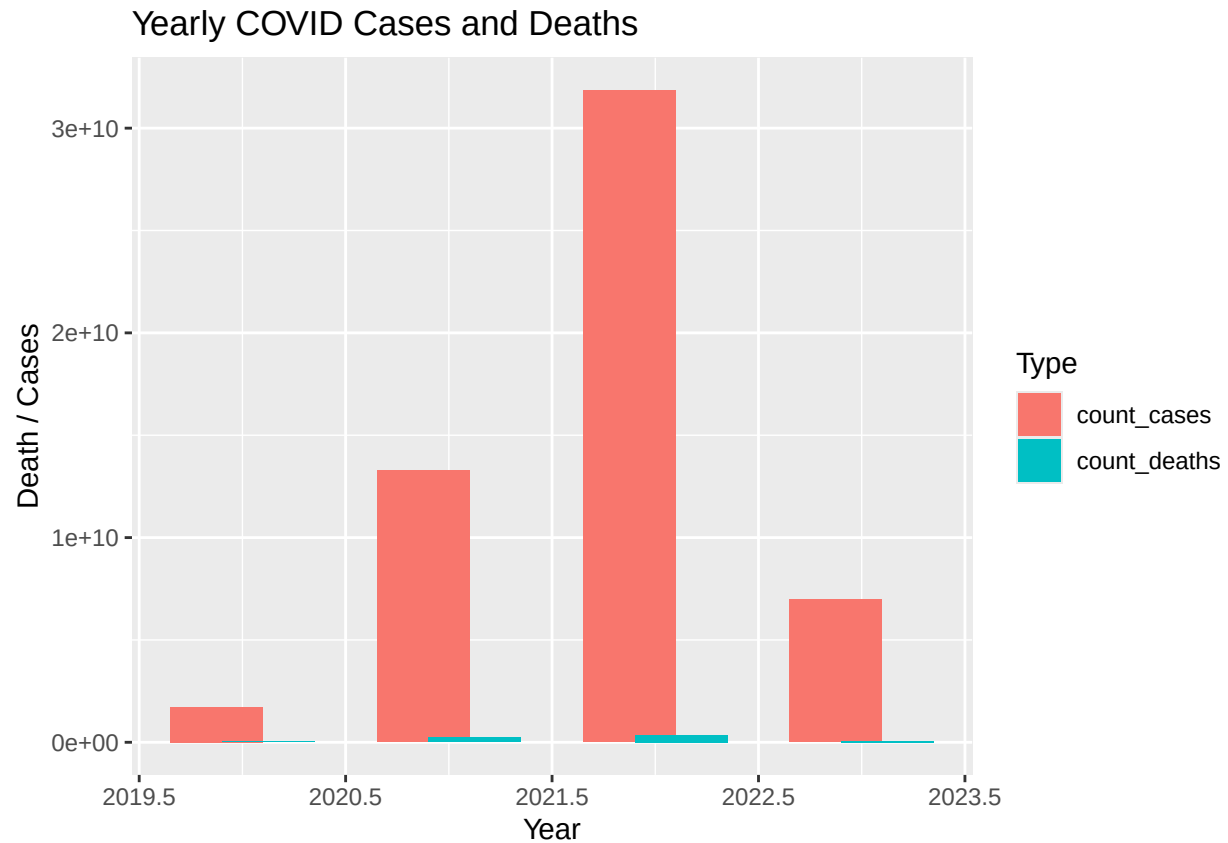
```
ggplot(yearly_counts, aes(x = yearly, y = count_cases)) +  
  geom_bar(stat = "identity", fill = "red") +  
  labs(title = "Yearly COVID Cases", x = "Year", y = "Count CASES")
```



```
ggplot(yearly_counts, aes(x = yearly, y = count_deaths)) +  
  geom_bar(stat = "identity", fill = "blue") +  
  labs(title = "Yearly COVID Deaths", x = "Year", y = "Count DEATHS")
```



```
yearly_long <- yearly_counts %>% pivot_longer(!yearly, names_to = "Type", values_to = "Count")  
  
ggplot(yearly_long, aes(x = yearly, y = Count, fill = Type)) +  
  geom_bar(position = position_dodge(0.5), stat = "identity") +  
  labs(title = "Yearly COVID Cases and Deaths", x = "Year", y = "Death / Cases")
```

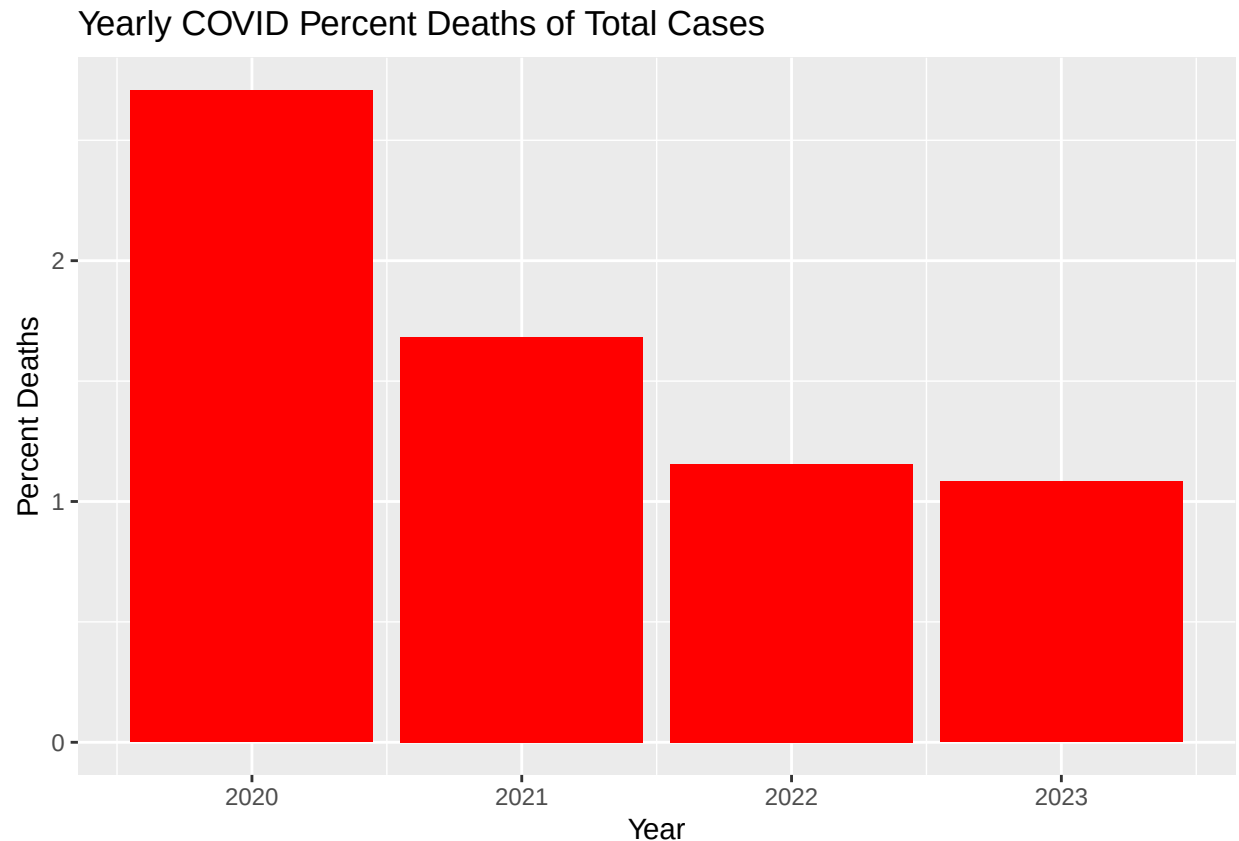


CREATED COLUMM FOR PERCENTAGE OF DEATHS

I then added a column in yearly counts to determine the percentage of deaths in relation to the total number of cases. I thought that this would help determine how effectively we are handling treatment of the cases over time and whether we are seeing increased immunity and caregiving. I then plotted this percentage of deaths in a bar chart for visualization.

```
yearly_counts <- yearly_counts %>%
  mutate(percent_deaths = (count_deaths)*100/count_cases)

ggplot(yearly_counts, aes(x = yearly, y = percent_deaths)) +
  geom_bar(stat = "identity", fill = "red") +
  labs(title = "Yearly COVID Percent Deaths of Total Cases", x = "Year", y = "Percent Deaths")
```

MODELING DEATH PERCENTAGE YEAR OVER YEAR

Finally, I created a linear model to predict the percentage of deaths in relation to the total cases over the years, to help see whether the trend was moving down over time. After analyzing the data and creating a visualization showing the year over year percentage and prediction, it became clear that the overall percentage of deaths is trending downwards, which is a positive sign for our prevention and handling of the COVID19 disease.

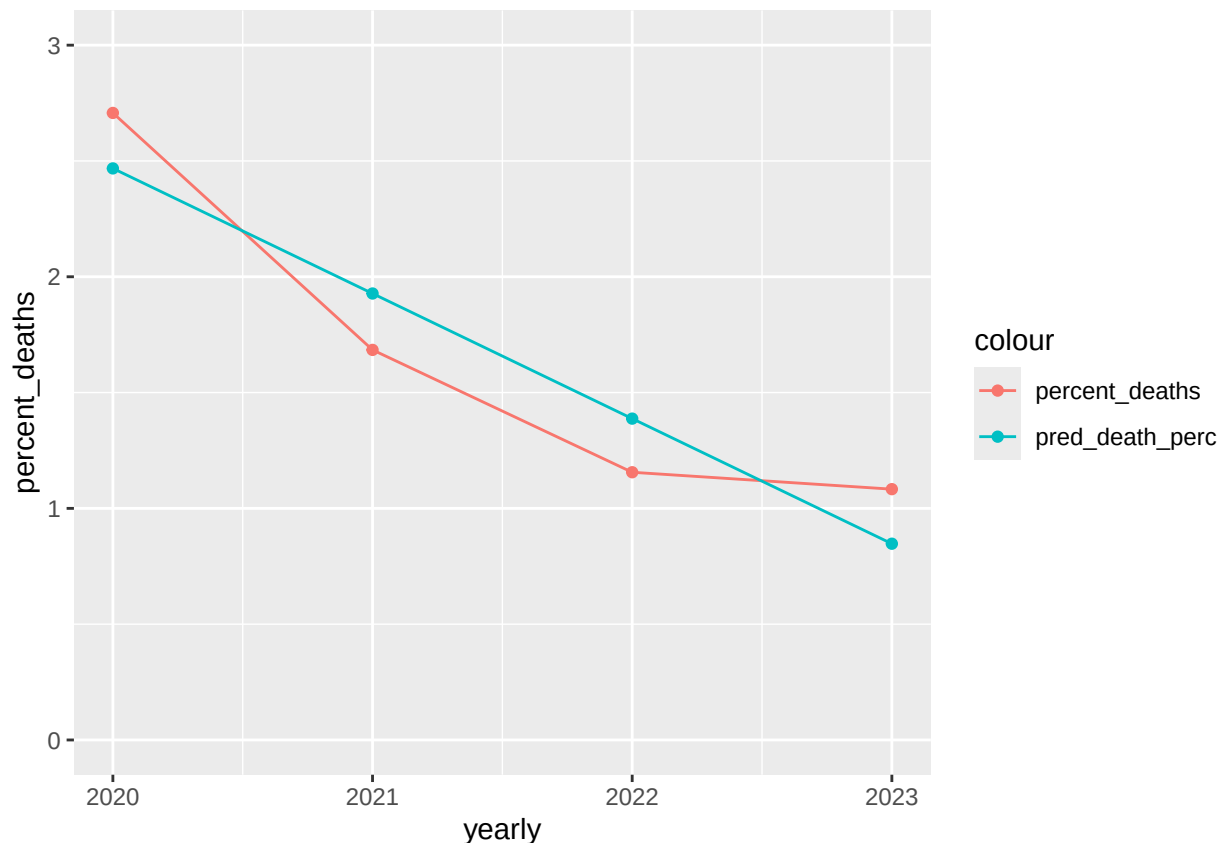
```
mod <- lm(percent_deaths ~ yearly, data = yearly_counts)
summary(mod)
```

```
##
## Call:
## lm(formula = percent_deaths ~ yearly, data = yearly_counts)
##
## Residuals:
##      1      2      3      4
## 0.2395 -0.2435 -0.2316  0.2356
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) 1093.5980   303.7424   3.600  0.0692 .
## yearly       -0.5402    0.1503  -3.595  0.0694 .
## ---
```

```
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.336 on 2 degrees of freedom
## Multiple R-squared:  0.866, Adjusted R-squared:  0.799
## F-statistic: 12.92 on 1 and 2 DF,  p-value: 0.06942
```

```
yearly_counts <- yearly_counts %>% mutate(pred_death_perc = predict(mod))

yearly_counts %>%
  ggplot(aes(x = yearly, y = percent_deaths)) +
  geom_line(aes(color = "percent_deaths")) +
  geom_point(aes(color = "percent_deaths")) +
  geom_line(aes(y = pred_death_perc, color = "pred_death_perc")) +
  geom_point(aes(y = pred_death_perc, color = "pred_death_perc")) +
  coord_cartesian(ylim = c(0,3))
```



CONCLUSION

Through my investigations in the COVID 19 pandemic, I learned about the overall trend of the disease globally, in the US, and state specific. The Johns Hopkins data set was very thorough, and I was able to manipulate the initial part of the data set through Dr. Jane Wall's guidance in the course lectures to create foundational analysis regarding the data. It was clear that the number of cases trended up exponentially as the disease spread, and that individual state saw a similar spread as the US as a whole. It was also very interesting to analyze the states that were impacted the most and the least throughout the years of the

pandemic, and understand how the new cases and new deaths were showing improvement in our treatment of the disease.

The additional analysis and visualizations I created focused on the yearly COVID19 cases and deaths, and the percentage of deaths in relation to the total number of cases. This helped to illuminate how deadly the disease was overall, which would indicate the severity, treatment practices, immunity, and other factors that could lead to a more stable spread. It was determined through modeling that the percent of deaths is trending downwards over the years, which would indicate a strong response to COVID19 and the ability for people to interact again, while taking precautions to not unnecessarily spread the disease.

I held several biases in the investigations of this data that I would like to address. Firstly, I was concerned about the overall severity of the disease and felt that it didn't match the social constraints placed on people at the time. That led me to question the death rate as a percentage, and to understand which states experienced the most deaths over time. It became clear that the percentage of deaths as 1/100 is rather high for any highly transmittable disease, which was helpful in my understanding of the actions put in place during the pandemic. Additionally, I was potentially biased in my search for reasons why the pandemic went on so long through a yearly understanding of the data. I graduated undergrad in the spring of 2020, and personally felt constrained at what could have been a very opportunistic time of life. This research helped to mitigate certain biases regarding my understanding of the severity of the disease overall and the reasons for lockdown.

Thank you ~ Frankie Carsonie ~ 10/1/2025